USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [23CSBS101]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

FUNDAMENTALS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Define variable. Give an example	L1	CO1	**02**
	b.	Find the value of the given expressions k = a*2+b/5-c*b where a=4, b=6 and c =1	L1	CO2	**02**
	c.	List out any four preprocessor directives in C.	L1	CO3	**02**
	d.	Define Array in C. Give an example	L1	CO4	**02**
	e.	Define recursive function in C	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain Structure of C program with examples	L2	CO1	**05**
		OR			
	b.	Explain Various data types in C program with example.	L2	CO1	**05**
3.	**a.**	Write a C program to find the factorial of n.	L3	CO2	**05**
		OR			

	b.	Find the output of the following programs: Explain it
		main( )
		{
		int k, num = 30 ;
		k = (num > 5 ? (num <= 10 ? 100 : 200) : 500) ;
		printf ("\n%d", num) ;
		}
			L3	CO2	**05**

4.	**a.**	Explain different types of Unary Operators in C.	L2	CO2	**05**
		OR			
	b.	Determine the hierarchy of operations and evaluate the following expression: i = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8	L3	CO2	**05**

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5.	**a.**	Discuss in detail about the preprocessor in C programming language	L2	CO3	**05**
		OR			
	b.	Explain the scope of global and local variable using example?	L2	CO3	**05**
6.	**a.**	Write a C program to swap the contents of two variables using pointer	L3	CO4	**05**
		OR			
	b.	Discuss in detail about the pointer in C with example	L2	CO4	**05**
7.	**a.**	Define Structure data type in C. Explain its declaration and initialization with example	L2	CO5	**05**
		OR			
	b.	Define Union data type in C. Explain its declaration and initialization with example	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss in detail about the Formatted input and output statement in C .	L2	CO1	**10**
		OR			
	b.	Discuss in detail about the different types of Constants and Variables in C? Explain Rules for Constructing Variable Names with an example	L2	CO1	**10**
9.	**a.**	Explain different types of Operators in C with examples	L2	CO2	**10**
		OR			
	b.	Describe the Conditional operator. Write a C program to find the given number is odd or even	L2	CO2	**10**
10.	**a.**	Describe the following C programming language statement with example C program **i.** Simple if Statement **ii.** if-else Statement **iii.** Nested if-else Statement	L2	CO2	**10**
		OR			
	b.	Explain the following looping statements in C with syntax and examples **i.** for statement **ii.** while statement **iii.** do-while statement	L2	CO2	**10**
11.	**a.**	Write a C program to Concatenate of two strings without using string manipulation function.	L3	CO3	**10**

		OR			
	b.	Discuss the concept of Pass by value and Pass by reference with example.	L2	CO3	**10**
12.	**a.**	Define Pointers in C Programming Language? Explain the advantages and disadvantages in a pointer .	L2	CO4	**10**
		OR			
	b.	Define an array. Explain the declaration and initialization of one dimensional and two dimensional array with example.	L2	CO4	**10**
13.	**a.**	Discuss in detail about the following file functions in C programming language. **i.** fopen() **ii.** fclose() **iii.** gets() **iv.** fputs()	L2	CO5	**10**
		OR			
	b.	Write a C program to implementation of student's mark statement using array structure.	L3	CO5	**10**